

4月10日. 周四. 第15次课.

Topic: 定积分的换元法与分部积分法.

使用之前讲述的不定积分方法, 结合微积分基本定理, 理论上来说我们已经能求很多定积分了.

求解定积分 $\int_a^b f(x) dx$ 的策略:

step 1. 求解不定积分 $\int f(x) dx$, 获得 $f(x)$ 的一个原函数 $F(x)$;

step 2. 利用微积分基本定理, 有 $\int_a^b f(x) dx = F(x)|_a^b = F(b) - F(a)$.

e.g. 1. 求定积分 $\int_1^e \frac{\ln x}{x} dx$.

step 1: $\int \frac{\ln x}{x} dx = \int \ln x \cdot d(\ln x) \stackrel{t=\ln x}{=} \int t dt = \frac{t^2}{2} + C = \frac{1}{2} (\ln x)^2 + C$.

$\therefore F(x) = \frac{1}{2} (\ln x)^2$ 是 $\frac{\ln x}{x}$ 的一个原函数.

step 2: $\int_1^e \frac{\ln x}{x} dx = F(e) - F(1) = \frac{1}{2} (\ln e)^2 - \frac{1}{2} (\ln 1)^2 = \frac{1}{2} \cdot 1 - \frac{1}{2} \cdot 0 = \frac{1}{2}$.

所以说, 只要能求出 $f(x)$ 的不定积分,

就能求出 $f(x)$ 的定积分.

一些技巧可以让定积分的计算过程比不定积分的简单一些.

以下我们将分别讨论定积分的第一换元法. 第二换元法与分部积分法.

(1). 定积分的第一换元法. (The Substitution Rule).

[中文课本: 6.1 定积分的换元法. P227~P228; Ca/culus: P416~P417]

如果被积函数 $u(x)$ 可以看成 $u(x) = f(g(x)) \cdot g'(x)$, (其中 f 连续, g 可导).

$$\text{则 } \int_a^b u(x) dx = \int_a^b f(g(x)) \cdot g'(x) dx = \int_a^b f(g(x)) \cdot d(g(x))$$

$$\stackrel{t=g(x)}{=} \int_{g(a)}^{g(b)} f(t) \cdot dt.$$

核心想法: 定积分在进行换元积分 $t = g(x)$ 时,

要将所有与 x 相关的部分. (积分上下界, 积分函数, 积分变量).

都换成与 t 相关的内容.

$$\int_a^b f(g(x)) \cdot d(g(x)) \xrightarrow{t=g(x)} \int_{g(a)}^{g(b)} f(t) \cdot dt$$

Diagram annotations:
 - x from a to b $\rightarrow$ x to b
 - t from $g(a)$ to $g(b)$
 - t to $g(b)$
 - t to $g(a)$

e.g. 1. $\int_1^e \frac{\ln x}{x} dx = \int_1^e \ln x \cdot d(\ln x) \xrightarrow{t=\ln x} \int_0^1 t \cdot dt = \left(\frac{t^2}{2}\right) \Big|_0^1 = \frac{1}{2} - \frac{0}{2} = \frac{1}{2}$

可以看到, 我们并没有求 $\int \frac{\ln x}{x} dx$, 在换元积分时省去了

不定积分的分部积分中将 $t = g(x)$ 代入到 $\int f(t) dt$ 积分结果的一步.

证明: 利用不定积分的分部积分 & 微积分基本定理证明.

已知 $u(x) = f(g(x)) \cdot g'(x)$

若 $\int f(t) \cdot dt = F(t) + c$. 即 $\frac{d}{dt} F(t) = f(t)$.

则 $\int u(x) dx = F(g(x)) + c$. 即 $\frac{d}{dx} F(g(x)) = u(x)$

$\therefore \int_a^b u(x) dx = [F(g(x))] \Big|_a^b = F(g(b)) - F(g(a))$

$= F(t) \Big|_{g(a)}^{g(b)} = \int_{g(a)}^{g(b)} f(t) dt$

e.g. 2. $\int_{-1}^2 x^3 \cos(x^4+2) \cdot dx = \int_{-1}^2 \cos(x^4+2) \cdot \frac{1}{4} d(x^4+2) \xrightarrow{t=x^4+2} \int_3^{18} \cos t \cdot \frac{dt}{4}$
 $= \left(\frac{1}{4} \sin t\right) \Big|_3^{18} = \frac{1}{4} \cdot (\sin 18 - \sin 3)$

e.g. 3. $\int_0^{\pi/4} \frac{\tan^6 x}{\cos^5 x} dx$

$t = \tan x, \quad dt = \frac{1}{\cos^2 x} dx$

$\frac{1}{\cos^5 x} = \frac{\sin^2 x + \cos^2 x}{\cos^5 x} = \tan^2 x + 1 = t^2 + 1$

$= \int_0^{\pi/4} \frac{\tan^6 x}{\cos^5 x} \left(\frac{dx}{\cos^2 x}\right) \xrightarrow{t=\tan x} \int_0^1 t^6 \cdot (t^2+1) dt = \int_0^1 (t^8 + t^6) \cdot dt$

$= \left(\frac{1}{9} t^9 + \frac{1}{7} t^7\right) \Big|_0^1 = \left(\frac{1}{9} + \frac{1}{7}\right) - (0+0) = \frac{1}{9} + \frac{1}{7} = \frac{16}{63}$

$$\sin 2x = 2 \cdot \sin x \cdot \cos x$$

Try it

$$\int_0^1 e^{-x^2} \cdot x \, dx = \frac{1}{2} - \frac{1}{2e}$$

$$\int_{-\pi}^{\pi} \sin x \cdot \sin 2x \, dx$$

$$= 2 \cdot \int_{-\pi}^{\pi} \sin^2 x \cdot \cos x \, dx$$

$$= 2 \cdot \int_{-\pi}^{\pi} \sin^2 x \cdot d(\sin x)$$

$$\stackrel{t = \sin x}{=} 2 \cdot \int_0^0 t^2 \cdot dt$$

$$= 2 \cdot \left(\frac{1}{3} t^3 \right) \Big|_0^0$$

$$= 0.$$

e.g. 4. $\int_0^2 \frac{dx}{1 + \sin x}$

$$t = \tan \frac{x}{2}. \quad \sin x = \frac{2t}{1+t^2}. \quad dx = \frac{2}{1+t^2} dt.$$

$$\begin{aligned} &= \int_0^{\tan 1} \frac{\frac{1}{1+t^2} dt}{1 + \frac{2t}{1+t^2}} = \int_0^{\tan 1} \frac{1 \cdot dt}{(t+1)^2} = \left(\frac{-1}{t+1} \right) \Big|_0^{\tan 1} = \frac{-1}{\tan 1 + 1} - \left(\frac{-1}{1} \right) \\ &= 1 - \frac{1}{\tan 1 + 1} \end{aligned}$$

(2). 定积分的第二换元法 (Inverse Substitution).

[中文课本: 6.1 定积分的换元法. P228~P229; Calculus: P487, P490]

对 $\int_a^b f(x) dx$. 若 $x = g(t)$, $g(\alpha) = a$, $g(\beta) = b$

且 $t \in [\alpha, \beta]$ 时 $x = g(t)$ 不超出 $f(x)$ 的定义域.

$$\text{则 } \int_a^b f(x) \cdot dx = \int_{\alpha}^{\beta} f(g(t)) \cdot g'(t) dt.$$

在定积分的第二积分法中, 我们不用求出 $f(x)$ 的原函数.

因此不需要将 $t = g^{-1}(x)$ 代入 $\int f(g(t)) \cdot g'(t) dt$ 积分结果中.

因此不求 $x = g(t)$ 可逆, 但要求验证 $x = g(t)$ 是否会超出 $f(x)$ 的定义域.

Hw!

证明: 留作习题.

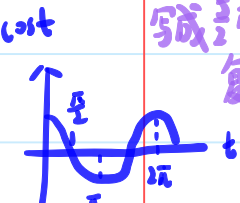
$$\cos^2 t = \frac{\cos 2t + 1}{2}$$

e.g. 5. $\int_0^a \sqrt{a^2 - x^2} dx \xrightarrow{x = a \cdot \sin t} \int_0^{\pi/2} a \cdot \cos t \cdot a \cdot \cos t \cdot dt = a^2 \int_0^{\pi/2} \cos^2 t dt$

积分上限也可以写成 $\frac{3\pi}{2}$, 但会复杂些.

$$= a^2 \int_0^{\pi/2} \frac{\cos 2t + 1}{2} dt = a^2 \left[\frac{1}{4} \sin 2t + \frac{t}{2} \right]_0^{\pi/2}$$

$$= a^2 \left[\left(\frac{0}{4} + \frac{\pi}{4} \right) - (0 + 0) \right] = \frac{\pi}{4} a^2.$$



$$= \int_0^{\pi/2} \sqrt{a^2 \cos^2 t} \cdot a \cdot \cos t dt = \int_0^{\pi/2} a \cdot \cos t \cdot a \cdot \cos t dt = \frac{\pi}{4} a^2.$$

$$+ \int_{\pi/2}^{3\pi/2} (-a \cdot \cos t) \cdot a \cdot \cos t dt$$

$$+ \int_{3\pi/2}^{5\pi/2} (a \cdot \cos t) \cdot a \cdot \cos t dt$$

e.g. 6. $\int_0^{\sqrt{3}/2} \frac{x^3}{(4x^2 + 9)^{3/2}} dx \xrightarrow{x = \frac{3}{2} \tan t} \int_0^{\pi/3} \frac{\frac{27}{8} \tan^3 t}{\left(9 \cdot \frac{1}{\cos^2 t} \right)^{3/2}} \cdot \frac{3}{2} \cdot \frac{dt}{\cos^2 t}$

$(t = \arctan \frac{2}{3} x)$.

$$= \int_0^{\pi/3} \frac{\frac{27}{8} \tan^3 t}{\frac{27}{\cos^3 t}} \cdot \frac{3}{2} \cdot \frac{dt}{\cos^2 t} = \int_0^{\pi/3} \frac{\frac{3}{8} \tan^3 t}{\frac{1}{\cos t}} dt = \int_0^{\pi/3} \frac{3}{16} \tan^2 t \cdot \cos t \cdot dt$$

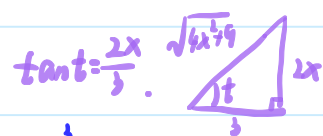
$$= \int_0^{\pi/3} \frac{3}{16} \tan^2 t \cdot \cos^3 t \left(\frac{\tan t}{\cos t} \right) dt \xrightarrow{u = \cos t} \int_1^2 \frac{3}{16} (u^2 - 1) \frac{1}{u^2} du.$$

$\tan^2 t + 1 = \frac{1}{\cos^2 t}$

$d(\frac{1}{\cos t}) = \frac{\tan t}{\cos^2 t} dt$

$$= \frac{3}{16} \cdot \int_1^2 \left(1 - \frac{1}{u^2}\right) \cdot du = \frac{3}{16} \cdot \left[u + \frac{1}{u}\right]_1^2$$

$$= \frac{3}{16} \cdot \left[\left(2 + \frac{1}{2}\right) - (1 + 1)\right] = \frac{3}{32}.$$



$$\int \frac{x^3}{(4x^2+9)^{3/2}} dx = u + \frac{1}{u} + C = \cos t + \frac{1}{\cos t} + C$$

$$= \frac{3}{\sqrt{4x^2+9}} + \frac{\sqrt{4x^2+9}}{3} + C.$$

Try it!

$$\int_0^3 \frac{x dx}{\sqrt{36-x^2}} \quad \underline{\underline{x=6 \cdot \sin t}} \quad \int_0^{\pi/6} \frac{6 \cdot \sin t}{6 \cdot \cos t} \cdot 6 \cdot \cos t dt$$

$$= 6 \cdot \int_0^{\pi/6} \sin t dt = 6 \cdot (-\cos t) \Big|_0^{\pi/6}$$

$$= 6 \cdot \left[-\frac{\sqrt{3}}{2} - (-1)\right] = 6 - 3\sqrt{3}.$$

$$\int_2^3 \frac{dx}{(x^2-1)^{3/2}} \quad \underline{\underline{x = \frac{1}{\cos t}}} \quad \int_{\pi/3}^{\arccos 1/3} \frac{1}{(\tan^2 t)^{3/2}} \cdot \frac{\tan t}{\cos t} \cdot dt.$$

$$= \int_{\pi/3}^{\arccos 1/3} \frac{1}{\tan^2 t} \cdot \frac{\tan t}{\cos t} dt = \int_{\pi/3}^{\arccos 1/3} \frac{1}{\tan^2 t} \cdot \frac{1}{\cos t} \cdot dt.$$

$$= \int_{\pi/3}^{\arccos 1/3} \frac{\cos t}{\sin^2 t} dt \quad \underline{\underline{u = \sin t}} \quad \int_{\sqrt{3}/2}^{\sin(\arccos 1/3)} \frac{du}{u^2} = \int_{\sqrt{3}/2}^{2\sqrt{2}/3} \frac{du}{u^2}$$

$\theta = \arccos \frac{1}{3} \quad \cos \theta = \frac{1}{3}$
 $\sin \theta = \sqrt{1 - \cos^2 \theta} = \frac{2\sqrt{2}}{3}$

$$= \left(-\frac{1}{u}\right) \Big|_{\sqrt{3}/2}^{2\sqrt{2}/3} = -\frac{3}{2\sqrt{2}} - \left(-\frac{2}{\sqrt{3}}\right)$$

$$= \frac{2\sqrt{3}}{3} - \frac{3\sqrt{2}}{4}$$

(3) 定积分的分部积分法. (Integration by Parts).

[中文课本: 6.2 定积分的分部积分法. P229~P232; Calculus: P475~P476]

$$\int_a^b u(x) \cdot v'(x) dx = u(x) \cdot v(x) \Big|_a^b - \int_a^b u'(x) v(x) dx.$$

Hint! 

证明: 直接使用微积分基本定理即可. 留作习题.

e.g. 7.
$$\begin{aligned} \int_1^2 x \cdot \ln x \cdot dx &= \int_1^2 \ln x \cdot \frac{d(x^2)}{2} = \left(\ln x \cdot \frac{x^2}{2} \right) \Big|_1^2 - \int_1^2 \frac{x^2}{2} \cdot d(\ln x) \\ &= 2 \ln 2 - 0 \cdot \frac{1}{2} - \int_1^2 \frac{x}{2} \cdot dx = 2 \cdot \ln 2 - \left(\frac{x^2}{4} \right) \Big|_1^2 \\ &= 2 \cdot \ln 2 - \left(\frac{4}{4} - \frac{1}{4} \right) = 2 \cdot \ln 2 - \frac{3}{4}. \end{aligned}$$

e.g. 8.
$$\begin{aligned} \int_0^1 \arctan x \cdot dx &= \left(x \cdot \arctan x \right) \Big|_0^1 - \int_0^1 x \cdot d(\arctan x) \\ &= 1 \cdot \frac{\pi}{4} - 0 \cdot 0 - \int_0^1 \frac{x}{1+x^2} dx = \frac{\pi}{4} - \int_0^1 \frac{\frac{1}{2} d(1+x^2)}{1+x^2} \\ &\stackrel{t=1+x^2}{=} \frac{\pi}{4} - \int_1^2 \frac{dt}{2t} = \frac{\pi}{4} - \left(\frac{1}{2} \ln |t| \right) \Big|_1^2 = \frac{\pi}{4} - \frac{1}{2} \ln 2. \end{aligned}$$